# Identifying Ocean Sound Sources

## Biological, anthropogenic and geophysical signals

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This Higher Education Teaching Material was supported by the EGU Higher Education Teaching Material Grant 2023.
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Source: NOAA

## Learning objectives

By the end of this unit, you should be able to:

- Identify broad hydroacoustic source categories.
- Compare waveform, spectrum and spectrogram characteristics.
- Use frequency, duration and temporal pattern to classify sounds.
- Recognise uncertainty in acoustic classification.

# Three broad source categories

| Biological | Anthropogenic | Geophysical / environmental |
|---|---|---|
| Whales | Ships | Earthquakes |
| Dolphins | Sonar | Volcanic activity |
| Fish | Pile driving | Waves |
| Snapping shrimp | Seismic surveys | Rain / sea ice |

# What features help identify a sound?

When analysing an unknown sound, consider several characteristics together:

- **Dominant frequency** — where is most of the energy?
- **Bandwidth** — narrowband or broadband?
- **Duration** — milliseconds, seconds or minutes?
- **Temporal structure** — continuous or impulsive?
- **Signal type** — tonal, noisy, pulsed?
- **Repetition** — does the pattern repeat?
- **Change through time** — does frequency or amplitude change?

No single feature normally identifies a source with certainty.

# Biological sounds

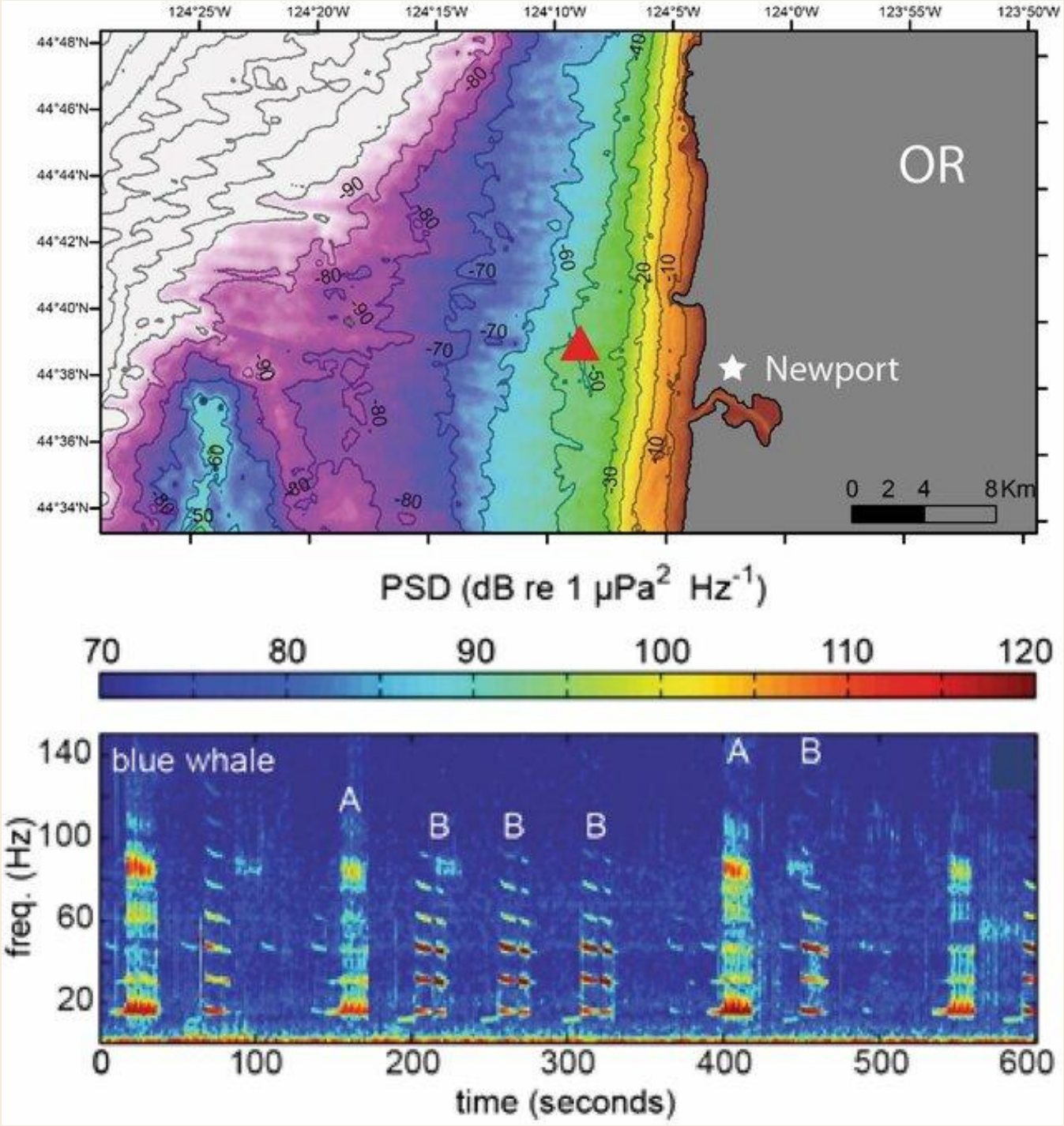
Marine animals produce sound for different functions:

- communication
- navigation
- echolocation
- locating prey
- mating and territorial behaviour

Different species and even different behaviours within the same species can produce very different acoustic signatures.

**Check the following sounds and try to identify the differences:**

https://www.oceannetworks.ca/multimedia/sounds/




Blue whale A and B calls recorded off Newport, Oregon. The upper panel shows the hydrophone location; the lower panel shows the recorded spectrogram. Source: Dziak et al. (2017), Figure 1, CC BY 4.0.

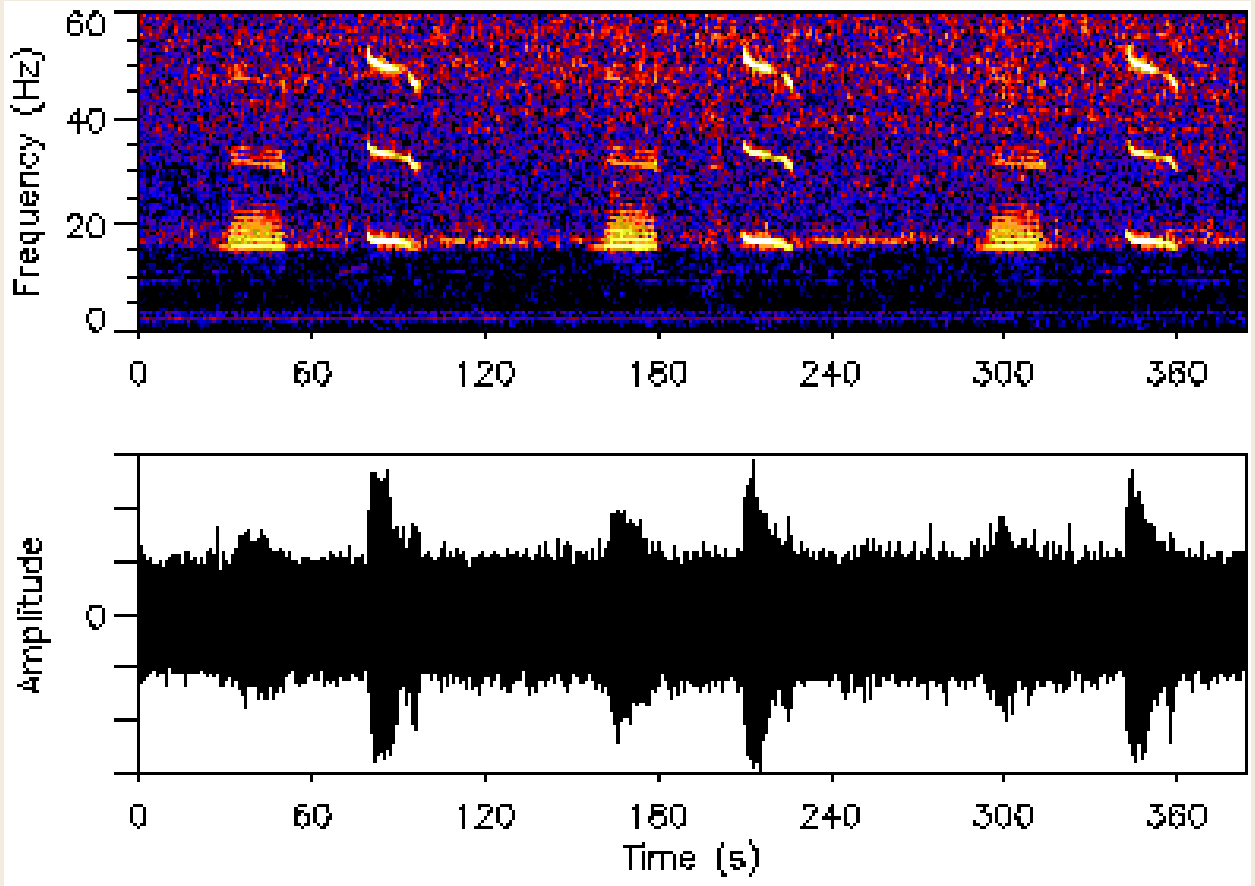
# Whale calls

Whale vocalisations vary strongly between species. Common features can include:

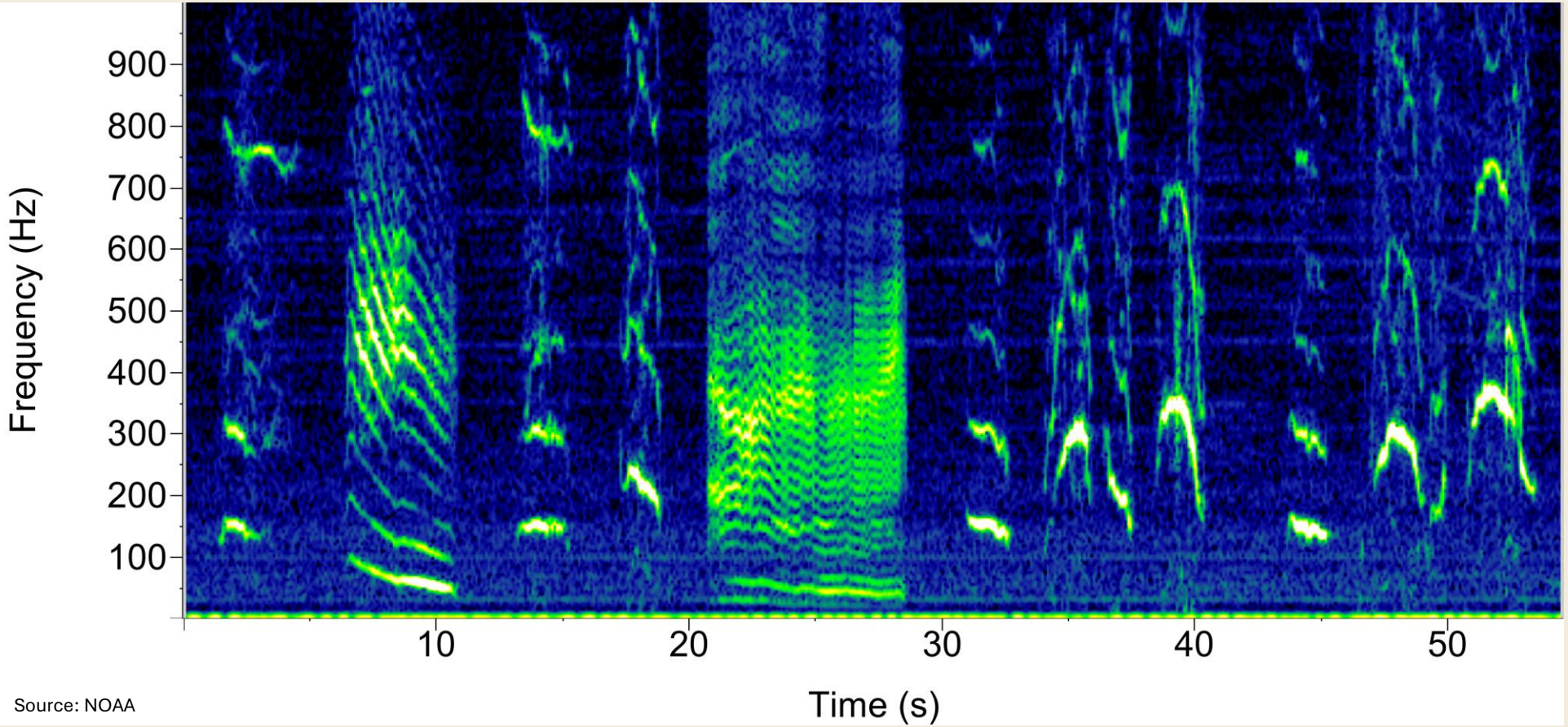
- tonal components
- structured or repeated sequences
- frequency modulation
- calls lasting seconds to minutes

The frequency range is species-dependent.

**Important:**
Not all whale sounds look or sound the same.




Example: Northeast Pacific blue whale
A calls typically contain repeated pulses, while B calls are more sustained tonal signals that change gradually in frequency.




Source: NOAA

**Example — Humpback whale**

Humpback vocalisations can contain:

- tonal components
- frequency-modulated calls
- repeated patterns and sequences
- sounds extending over seconds or longer

**Look at the spectrogram:**

Can you identify repeated structures or frequency changes?

What to look for: Curved traces indicate changes in frequency, while similar patterns appearing repeatedly suggest structured vocalisations.

## Dolphins and toothed whales

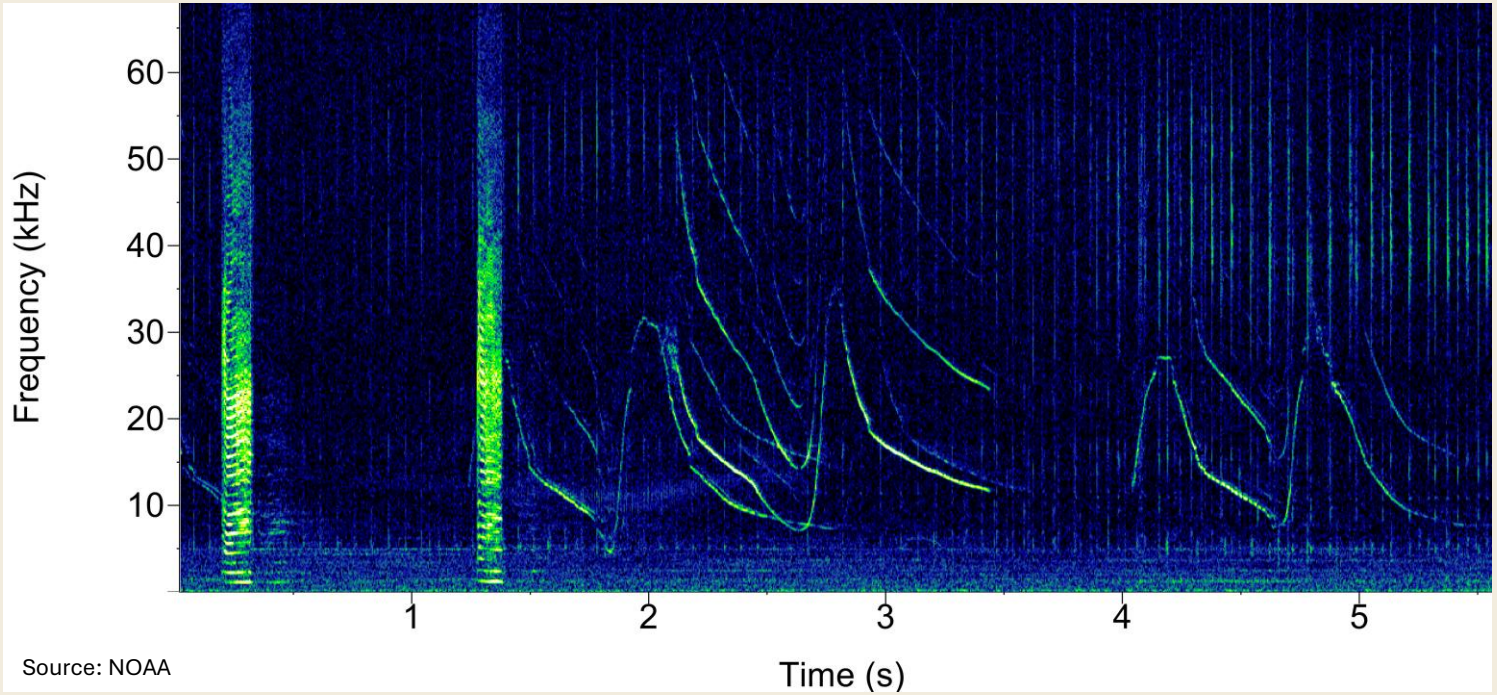
They can produce very different sound types.

### Whistles

- usually narrowband
- often frequency-modulated
- commonly associated with communication

### Echolocation clicks

- Very brief acoustic pulses.
- Often emitted in rapid sequences called click trains.
- Can be broadband or narrowband high-frequency, depending on the species.
- Used to detect objects, navigate and locate prey.



Source: NOAA

**Self-check:** Which would appear as a short vertical feature on a spectrogram: a whistle or a click?

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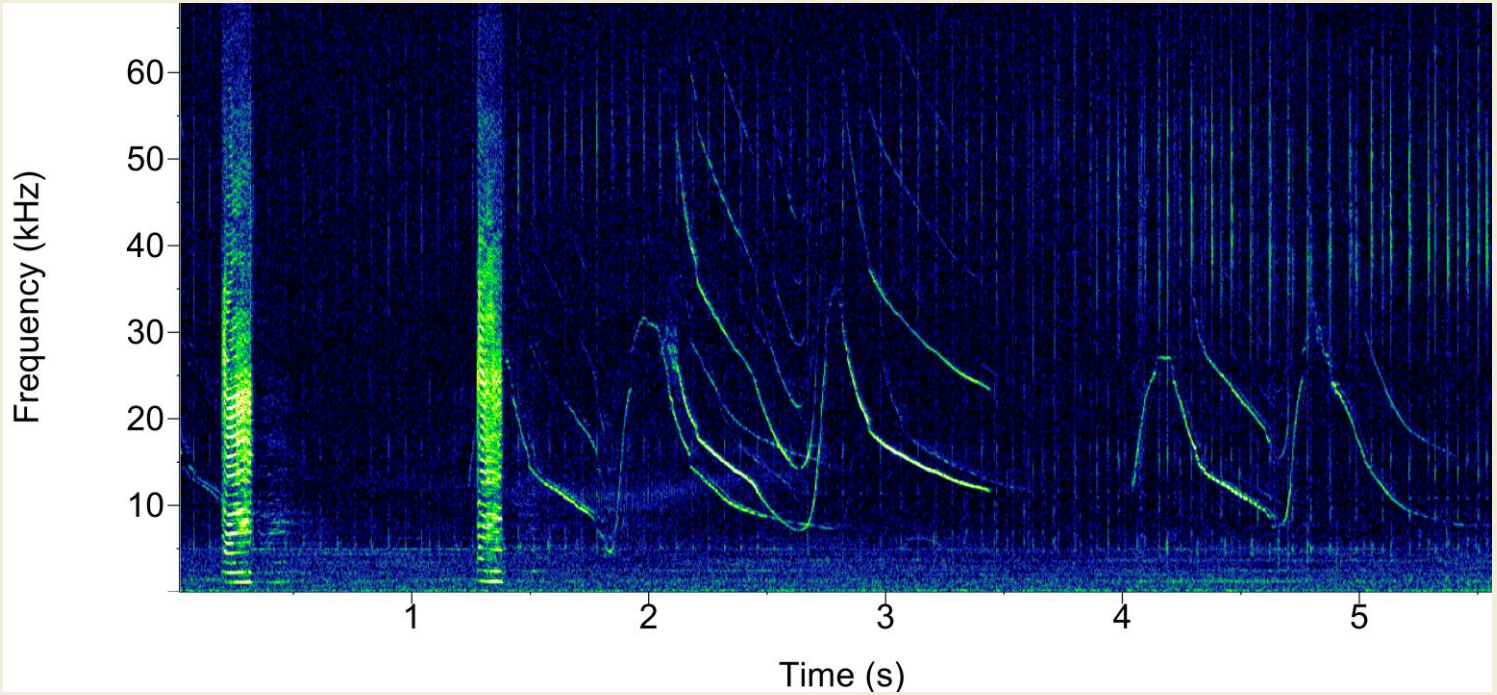
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Dolphin sound spectrogram showing short, broadband echolocation clicks appearing as vertical features and narrowband, frequency-modulated whistles appearing as curved traces. Source: NOAA Fisheries, *Sounds in the Ocean: Mammals*.

**Self-check:** Which would appear as a short vertical feature on a spectrogram: a whistle or a click?

**Answer:** click.

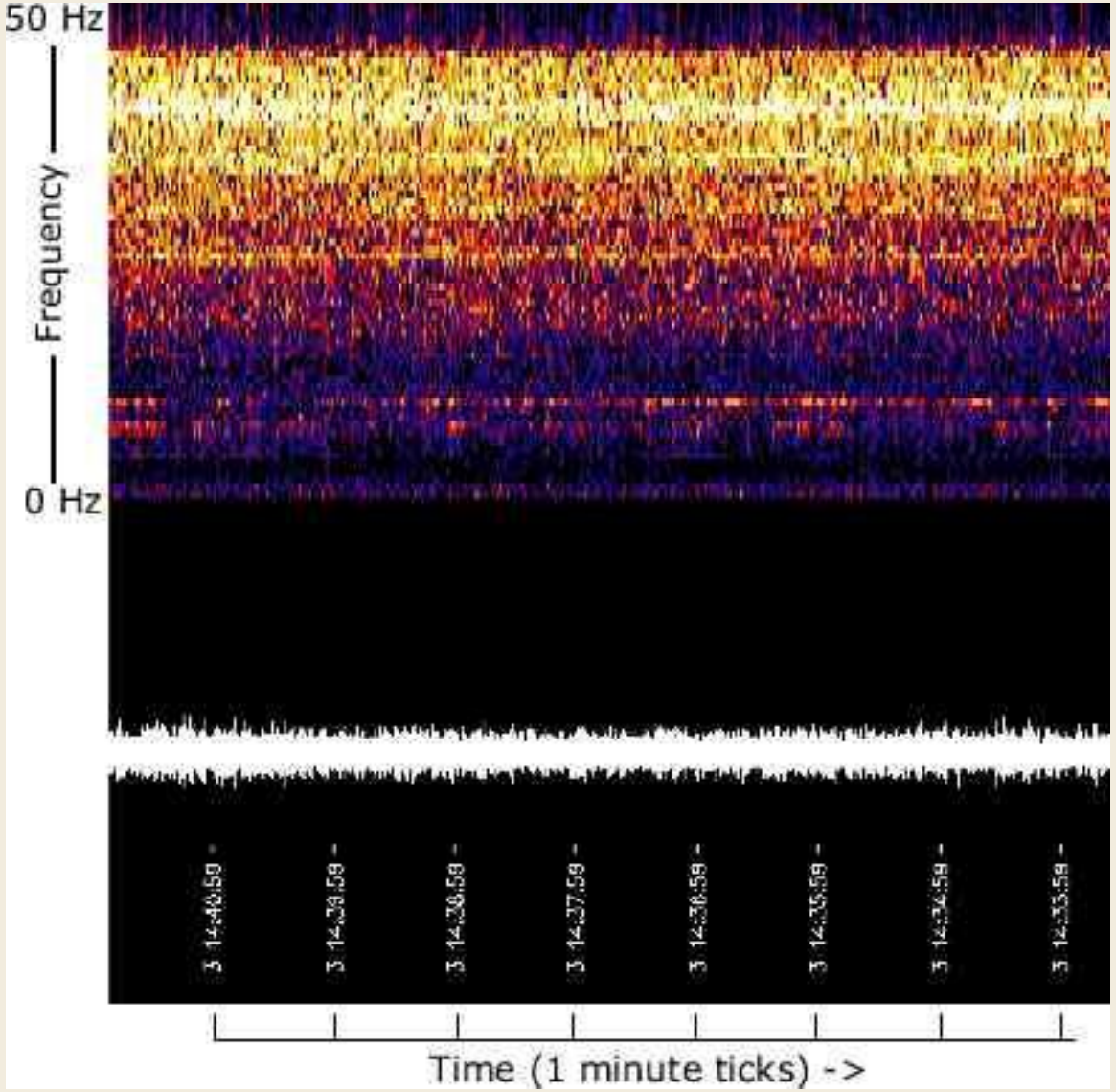
# Anthropogenic sound — ships

Common characteristics can include:

- relatively long-duration or continuous sound
- substantial low-frequency energy
- tonal components from machinery or propeller rotation
- broadband components associated with cavitation.

**Look at the spectrogram:** Does the signal persist through time? Are there repeated tonal bands? What can we identify? The sustained low-frequency energy and repeated tonal bands are characteristic of machinery and propeller-related vessel noise.

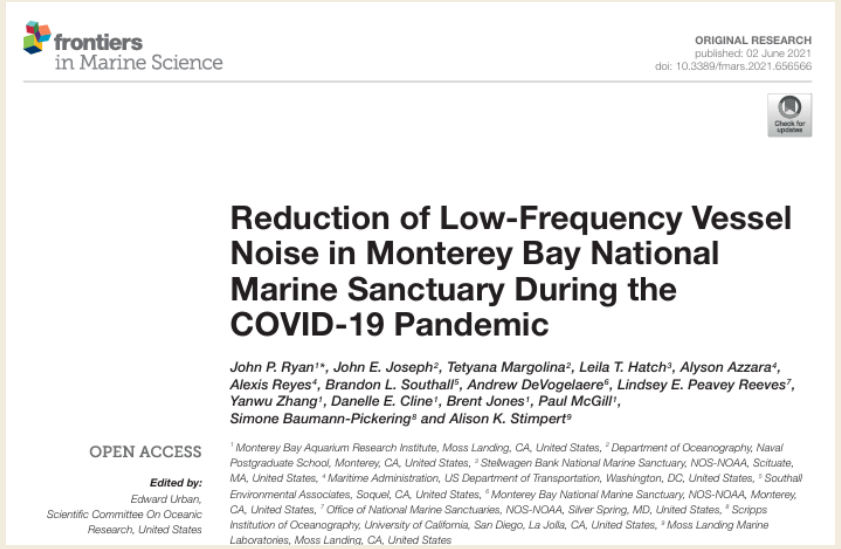
**Did you know?** Different vessels can produce characteristic acoustic signatures that may help scientists recognise vessel types and, under favourable conditions, individual ships.



# Real-world example: When shipping activity decreased, the ocean became quieter

During the COVID-19 pandemic, reduced vessel traffic provided an opportunity to study the contribution of shipping to underwater noise. Ryan et al. (2021) measured significant reductions in low-frequency ocean noise in Monterey Bay National Marine Sanctuary during 2020 compared with 2018–2019. In June 2020, mean noise spectrum levels in the 63 Hz one-third-octave band were approximately 1.9 dB lower than in previous years.

## What might reduced shipping noise mean for marine mammals that communicate using low-frequency sounds?



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published: 02 June 2021
doi: 10.3389/fmars.2021.656566

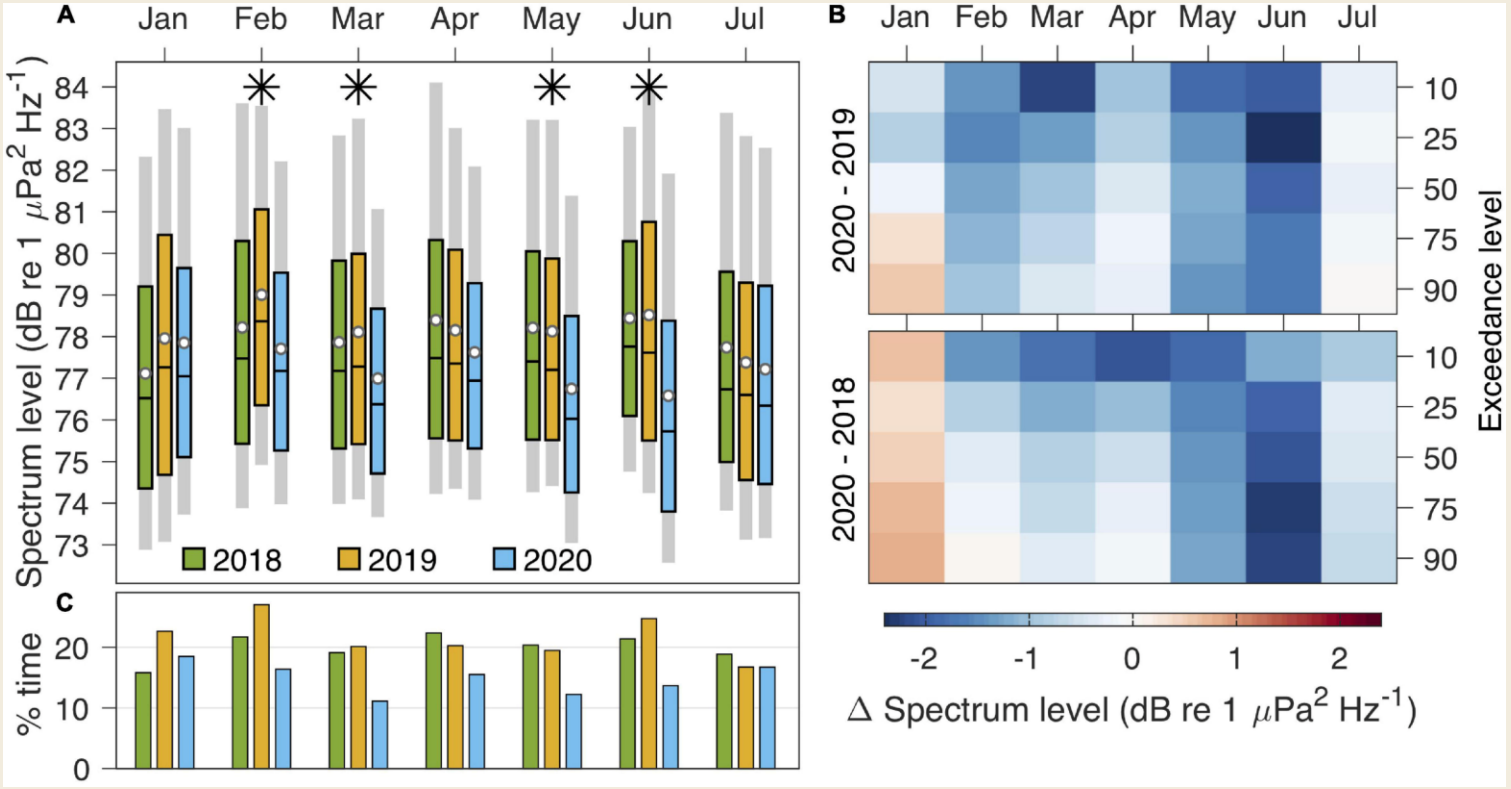
**Reduction of Low-Frequency Vessel Noise in Monterey Bay National Marine Sanctuary During the COVID-19 Pandemic**

*John P. Ryan[1]*, John E. Joseph[2], Tetyana Margolina[2], Leila T. Hatch[3], Alyson Azzara[4], Alexis Reyes[4], Brandon L. Southall[5], Andrew DeVogelaere[6], Lindsey E. Peavey Reeves[7], Yanwu Zhang[1], Danelle E. Cline[1], Brent Jones[1], Paul McGill[1], Simone Baumann-Pickering[8] and Alison K. Stimpert[9]*

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*Edited by:*
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Low-frequency underwater noise (one-third-octave band centred at 63 Hz) in Monterey Bay during 2018–2020. Noise levels were generally lower in 2020, particularly from February to June, consistent with reduced shipping activity during the COVID-19 pandemic.
Source: Ryan et al. (2021), Figure 6.

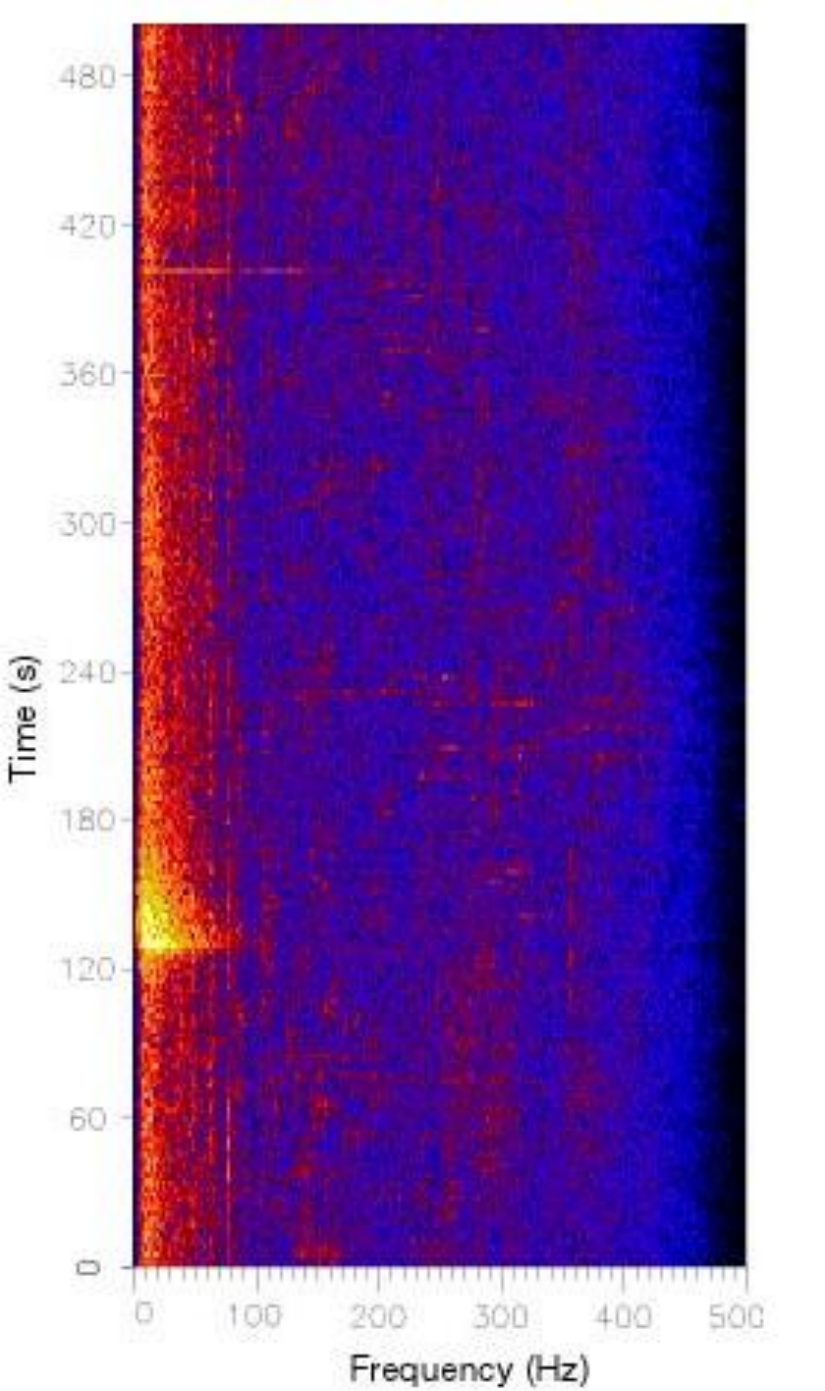
# Other human sources

| Source | Typical acoustic character |
|---|---|
| **Ship** | Long-duration / continuous |
| **Sonar** | Often tonal or pulsed |
| **Pile driving** | Repeated impulsive events |
| **Seismic survey** | Repeated broadband pulses |
| **Construction machinery** | Continuous or intermittent |

**Key distinction**
Human activities can generate both continuous and impulsive sound.

Optional real seismic-survey example:
**https://www.pmel.noaa.gov/acoustics/env-noise/env-noise.html**
The page includes an airgun spectrogram from seismic surveying.

**Example:** Hydroacoustic signal generated by an earthquake The NOAA recording contains a transient signal with strong low-frequency energy, particularly between approximately 5 and 30 Hz.




## Geophysical signals — Earthquakes

Submarine earthquakes can generate hydroacoustic signals that propagate through the ocean.

Typical characteristics may include:

- transient, event-like behaviour
- strong low-frequency energy
- broadband components
- complex waveforms
- long-distance propagation

Look at the spectrogram: Can you identify the short-lived acoustic event against the background noise?

Note: In this example, time is plotted vertically and frequency horizontally. The NOAA example on the left shows strongest acoustic energy approximately between 5 and 30 Hz.

# Compare the three

| **Feature** | Whale call | Ship | Earthquake |
|---|---|---|---|
| **Source category** | Biological | Anthropogenic | Geophysical |
| **Duration** | Seconds–minutes | Long / continuous | Transient |
| **Frequency** | Species-dependent | Often low-frequency dominated | Often low-frequency / broadband |
| **Pattern** | Structured / repeated | Persistent / tonal components | Event-like |
| **Typical clue** | Repeated vocal pattern | Continuous bands | Sudden complex event |

**Important**

These are broad tendencies, not strict identification rules.
Different source types can overlap in frequency and appearance.

## Environmental sound — ice cracking

Ice can generate acoustic signals through cracking, fracturing, calving and contact with the seafloor or other ice.

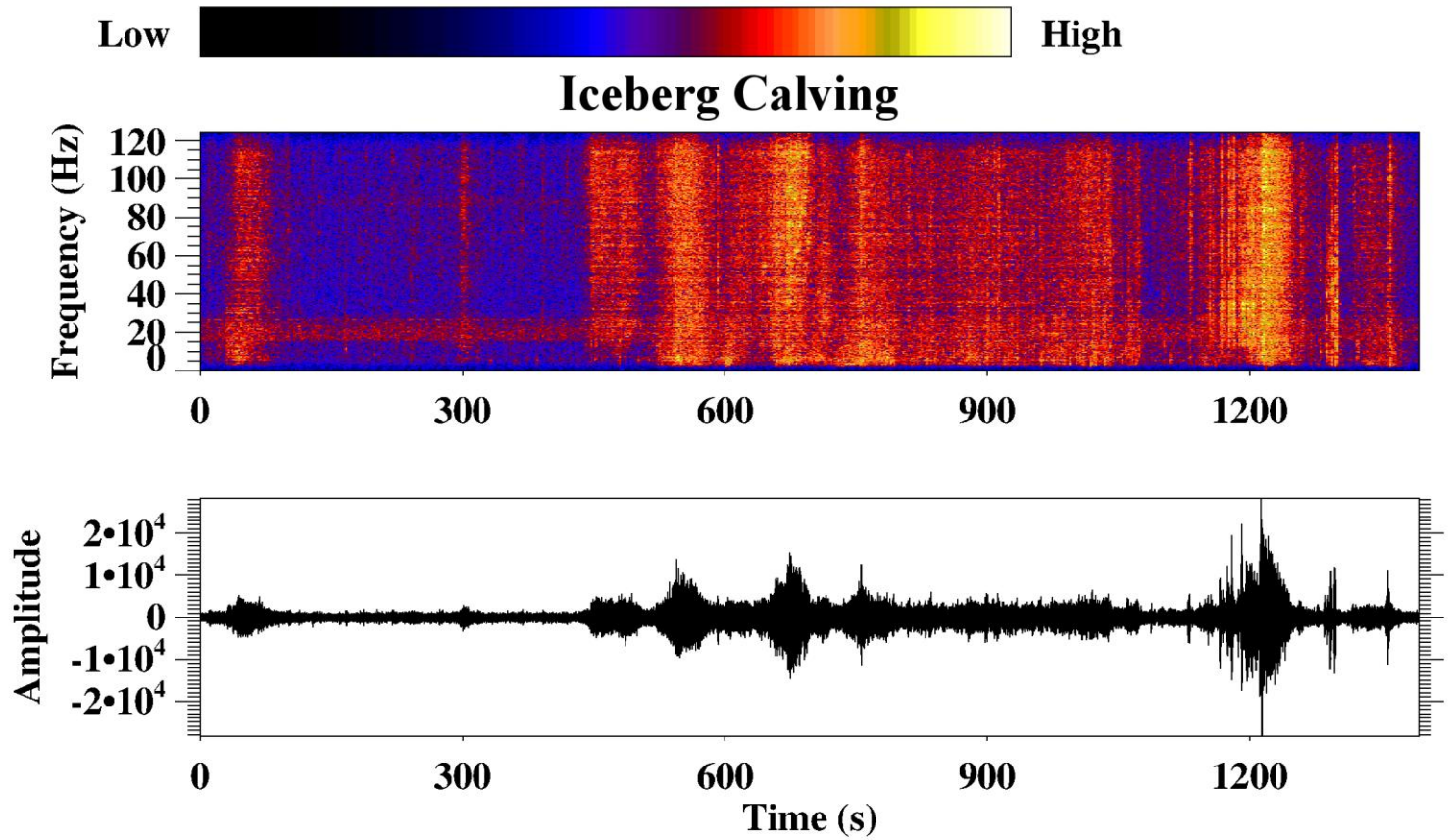
Ice-cracking signals can be:

- impulsive or transient
- broadband
- repeated during periods of active fracturing

**Look at the spectrogram:**

Is the signal tonal or broadband?

Does it occur as a continuous sound or as discrete events?




Spectrogram of an iceberg calving (large section of iceberg breaking off) while adrift. The calving signal is short duration, broad band from 1-440 Hz generated by ice cracking and crack propagation. Source: NOAA

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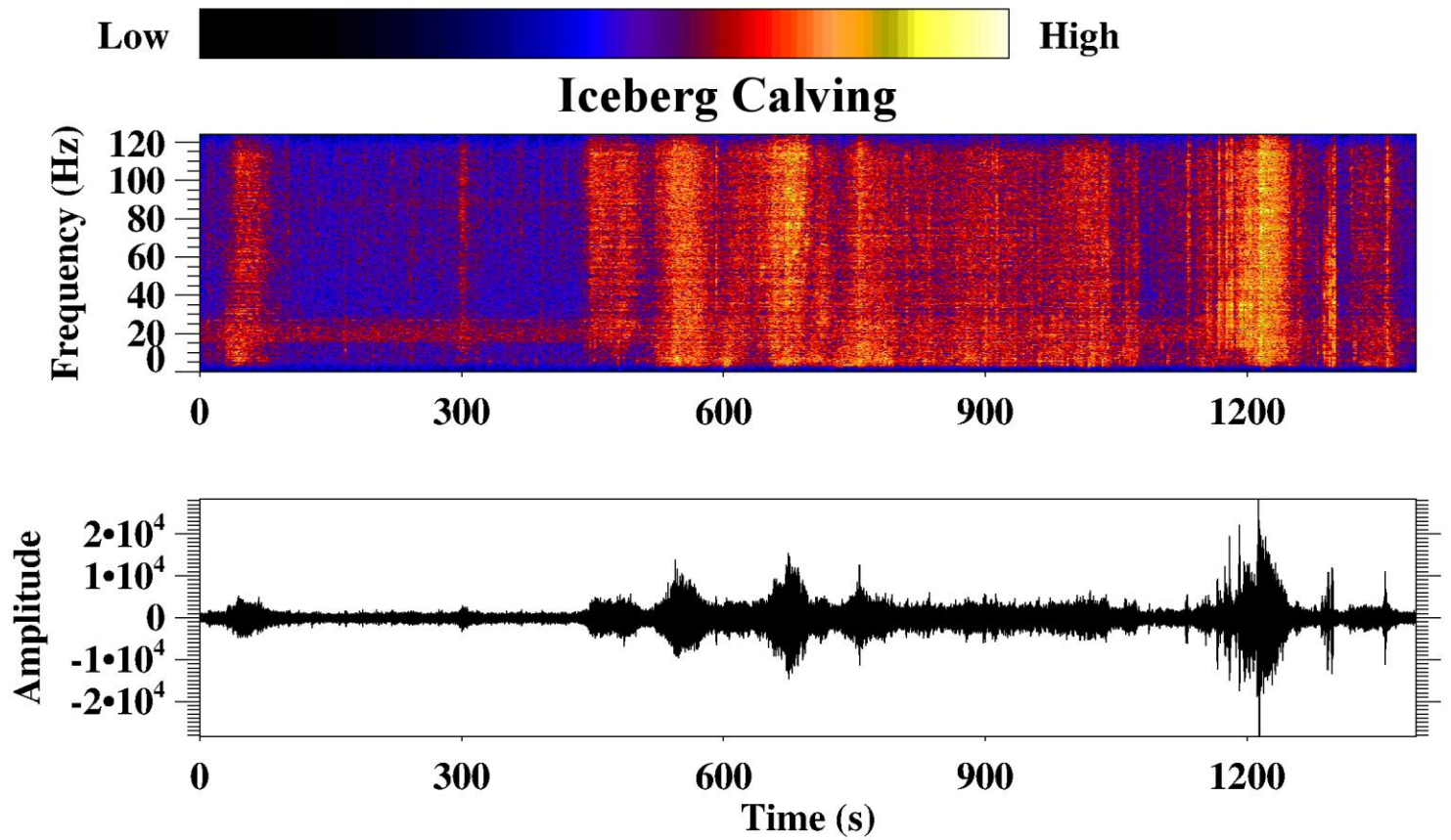
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**Interpretation:** The prominent acoustic events are broadband, with energy distributed across a range of frequencies rather than concentrated in sustained tones.
The recording contains several distinct periods of increased acoustic activity, including stronger events around 500–800 s and near 1,200 s. These are consistent with intermittent ice-fracture and calving activity.

# Why is classification not always easy?

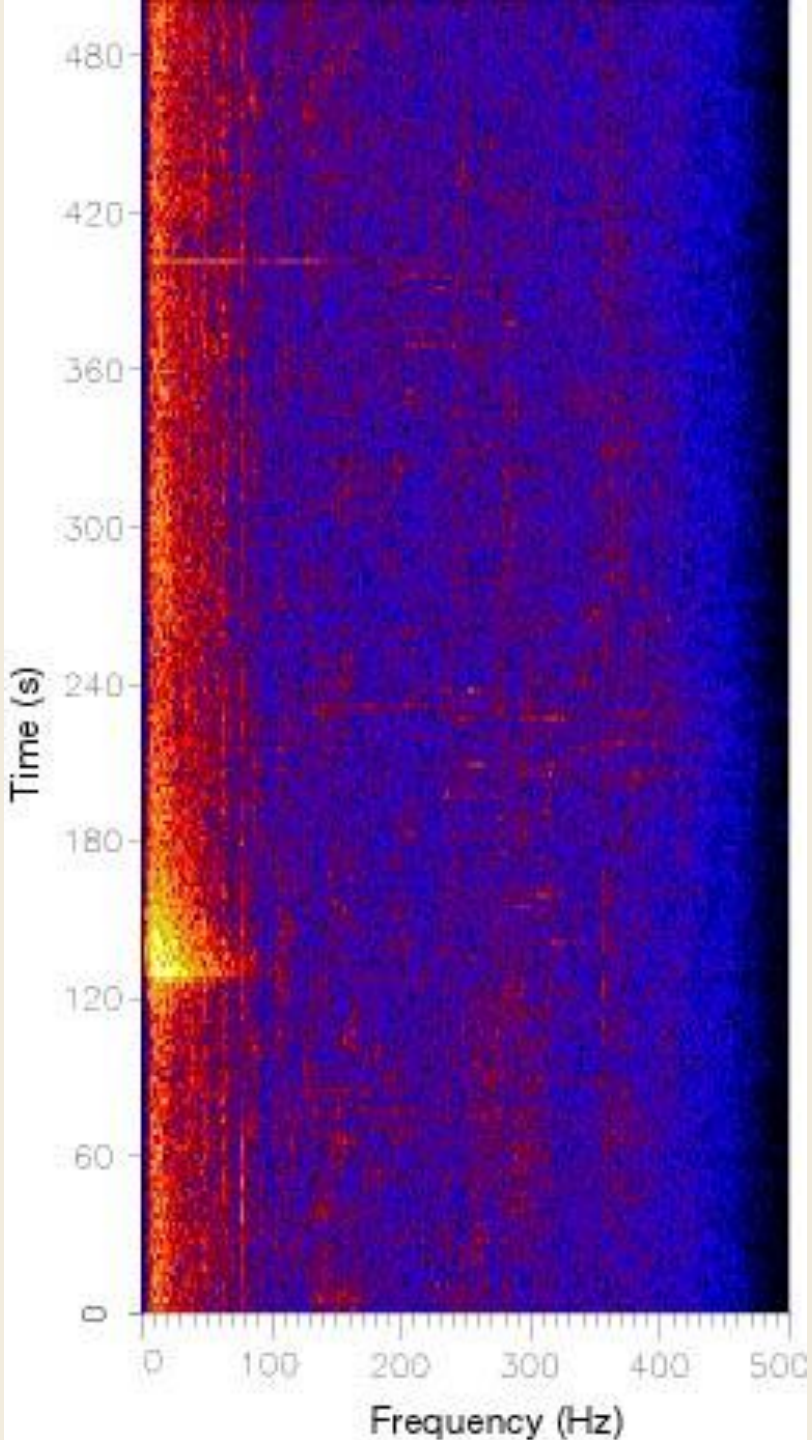
A recorded signal depends on more than the source itself. Classification can be affected by:

- overlapping frequency ranges
- distance from the source
- propagation effects
- background noise
- several sources occurring simultaneously
- low signal-to-noise ratio (SNR): The sound of interest is weak compared with the surrounding background noise, making it difficult to identify.

The sound recorded by a hydrophone depends on:

- The original acoustic source.
- How sound propagates from the source to the receiver.
- Other sounds occurring simultaneously.
- Ambient and instrument noise.

The same source may produce very different recordings depending on the environment and distance from the hydrophone.




This NOAA earthquake recording also contains humpback whale calls in the background. Listen to it here: **https://www.pmel.noaa.gov/acoustics/sounds/earthquake.html**

# Notebook 2 — Comparing and Classifying Ocean Sounds

You will analyse three underwater recordings:

- Humpback Whale Calls — biological
- Small Boat — anthropogenic
- Ice Cracking in Cambridge Bay — environmental

For each recording you will:

**load → waveform → FFT → spectrogram → describe features → classify**

You will compare frequency, bandwidth, duration, temporal pattern and signal character, then justify your classification and state your confidence.

**The goal is not simply to guess the answer, it is to explain what acoustic evidence supports your interpretation.**

## Take-home messages

- Ocean sounds can have biological, anthropogenic or geophysical origins.
- Frequency, bandwidth, duration and temporal structure are useful classification features.
- Waveforms, spectra and spectrograms provide complementary information.
- No single characteristic uniquely identifies a source.
- Propagation and background noise can modify the recorded signal.
- Acoustic classifications should include a statement of confidence or uncertainty.

# Explore further

**NOAA PMEL Acoustics Program**

Spectrograms and examples of marine mammals, ships, earthquakes and other underwater sounds.
https://www.pmel.noaa.gov/acoustics/sounds.html

**Ocean Networks Canada**

Archived hydrophone recordings from Pacific and Arctic observatories.
https://www.oceannetworks.ca/multimedia/sounds/

# References


**Scientific publications**

- Dziak, R. P., et al. (2017). A pulsed-air model of blue whale B call vocalizations. *Scientific Reports, 7*, 9122. https://doi.org/10.1038/s41598-017-09423-7
- Ryan, J. P., et al. (2021). Reduction of Low-Frequency Vessel Noise in Monterey Bay National Marine Sanctuary During the COVID-19 Pandemic. *Frontiers in Marine Science, 8*, 656566. https://doi.org/10.3389/fmars.2021.656566

**Scientific and educational resources**

- NOAA PMEL. *Marine Mammal and Environmental Sounds*. https://www.pmel.noaa.gov/acoustics/sounds.html
- NOAA Fisheries. *Sounds in the Ocean: Mammals*. https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-mammals
- Ocean Networks Canada. *Underwater Sound Recordings*. https://www.oceannetworks.ca/multimedia/sounds/

**Figure and media credits**

- Slide 1: Humpback whale photograph. NOAA.
- Slide 5: Blue whale calls and hydrophone location. Dziak et al. (2017), Figure 1, CC BY 4.0.
- Slide 6: Northeast Pacific blue whale spectrogram and waveform. NOAA PMEL Acoustics Program.
- Slide 7: Humpback whale song spectrogram. NOAA Fisheries, Northeast Fisheries Science Center, Passive Acoustics Branch.
- Slides 8–9: Dolphin click and whistle spectrogram. NOAA Fisheries, Passive Acoustics Branch.
- Slide 10: Ship noise spectrogram. NOAA PMEL, *NOAA Ship Ronald H. Brown*.
- Slide 11: Vessel noise spectrogram and waveform. Ryan et al. (2021), Figure 4, CC BY 4.0.
- Slides 13 and 17: Earthquake hydroacoustic spectrogram. NOAA PMEL Acoustics Program.
- Slides 15–16: Iceberg-calving spectrogram and waveform. NOAA PMEL Acoustics Program.